Syed Albash Ali

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EDUCATION

Ramdeobaba University, Nagpur, Maharashtra, India — Aug 2024 to May 2028 (Expected)
Bachelor of Technology in Electronics and Computer Science

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, Deep Learning, Embedded Systems, Cloud Computing, Linear Programming

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, Rust
Web Development: HTML, CSS, React.js, Vue.js, Node.js, REST APIs
Databases: SQL, NoSQL
AI/ML: PyTorch, LightGBM, Sentence Transformers, FAISS, NumPy, Natural Language Processing, Large Language Models (LLMs), Approximate Nearest Neighbor Search
Data Formats: Parquet
Tools & Platforms: Linux, Git, GitHub, Docker, VS Code, MATLAB
Embedded & IoT: Arduino, RFID, Weight Sensors

PROJECTS

Amazon ML Challenge 2026 – Multi-Source Entity Resolution 2026
Python, PyTorch, Sentence Transformers, FAISS, LightGBM, Parquet

- Built an end-to-end entity matching pipeline to resolve noisy business records across three large data sources containing **12.5M+ records** with names, addresses, and country information.
- Generated multilingual embeddings using **paraphrase-multilingual-MiniLM-L12-v2** and implemented **FAISS OPQ+IVFPQ approximate nearest neighbor search** for scalable candidate retrieval across millions of records.
- Engineered memory- and time-efficient processing using **Parquet streaming, FP16 embeddings, tensor caching, and training/validation caps**, achieving **0.999745 LightGBM training AUC** and a **0.66440 validation score**.

RFID & IoT-Based Smart Shopping Cart 2025
Python, Arduino, RFID, IoT, Weight Sensors

- Developed an automated retail system integrating RFID and weight sensors for real-time product identification, verification, and automated billing.
- Implemented real-time billing and in-store product navigation features to streamline the shopping and checkout experience.

Online Library Management System 2024
JavaScript, Python, HTML/CSS, Backend APIs

- Built a full-stack platform for authentication, book browsing, renting/downloading, inventory management, and borrowing records.
- Developed modular RESTful backend APIs to integrate frontend and backend systems and manage application data.

CERTIFICATIONS

Data-Driven Animation for Science Communication Coursera, 2024

- Applied data visualization techniques to communicate complex technical information through interactive media.

Interactive Computer Graphics Coursera, 2025

- Studied rendering pipelines, shader programming, and real-time graphics concepts.

Programming in Python (Meta) Coursera, 2026

- Covered Python fundamentals including OOP, error handling, testing, and libraries.

HTML and CSS in Depth (Meta) Coursera, 2026

- Built responsive layouts using modern CSS and front-end debugging techniques.

Java: Linear Data Structures and Trees (Codio) Coursera, 2026

- Implemented linked lists, stacks, queues, binary trees, and self-balancing trees.